6

Analyzing the Current Database

To see what is in front of one's nose
needs a constant struggle.
—GEORGE ORWELL
IN FRONT OF YOUR NOSE

Topics Covered in This Chapter

Getting to Know the Current Database

To determine where you should go, you must first understand where you are.

This maxim defines the entire philosophy behind this phase of the database design process. You must devote some time to gaining a clear understanding of your organization's database so that you can

- Determine whether the database supports the organization's *current* information requirements
- Uncover existing structural deficiencies
- Determine how the database needs to evolve so that it will support the organization's *future* information requirements

You can use the existing database as a resource for developing a new database. However, you must carefully judge which aspects of the current database remain useful and which aspects should be discarded. You can make these judgments by answering the following questions:

What types of data does the organization use?

How does the organization use that data?

How does the organization manage and maintain that data?

The answers to these questions provide you with vital information that you can use to design a database that best suits your organization's needs.

You can best answer these questions by analyzing your organization's existing database. It's very likely that the organization is using some type of database, and it can probably be associated with one of the following categories.

- *Paper-based databases.* Also known as *file systems*, these typically consist of various forms and handwritten or printed documents stored in file folders or bound in notebooks. The folders and notebooks are identified by some coding scheme (for example, unique numbers or colored tabs) and stored in file cabinets. These cabinets are likely to be identified by some coding scheme as well, depending on the size of the database. (You would be surprised how many businesses still have some form of paper-based system.)

- *Legacy databases* have been in existence and in use for several years and consist of various types of data structures and user interfaces that all reside on mainframe computers, network servers, personal computers, or more recently, on the cloud. The capability, functionality, and effectiveness of the structures and screens are quite dependent upon the skills and knowledge of the developers, the application development tools, and the database management software used to create them.
- *Human knowledge bases* (loosely defined) are based on the memory of one or more employees within an organization. These individuals have a specific amount of knowledge regarding a given aspect of the organization (for example, customer information or product details), and they are crucial to conducting the organization's business.

The goals of your analysis are to determine the types of data the organization uses, how the organization manages and maintains that data, and how the organization views and uses the data. You can reduce the time it takes to define the preliminary field and table structures for the new database if you conduct this investigation properly.

During the analysis, you review the various ways the organization collects and presents its data, and you conduct a set of interviews with users and management. You then use the information you've gathered to define a Preliminary Field List and to help determine the tables that should be included in the initial database structure. If your analysis reveals that the current database is poorly designed, you can take precautions to ensure that you don't make the same mistakes in the new database. Despite whatever shortcomings the current database may have, it can still help you identify a number of the fields and tables that you should include in the new database.

There's one rule you should keep first and foremost in your mind as you're analyzing the current database:

> *Do not adopt the current database structure as the basis for the new database structure.*

Following this rule will help you avert unnecessary errors and aid in maximizing your design efforts.

Every so often, there's a point during the analysis when a novice database developer (and sometimes an experienced one) will stop and think, "This database doesn't look too bad. Let's just end the analysis here and use this database as the basis for the new one." This is a particularly bad idea because every hidden problem within the current database structure will be transferred into the new database. These types of problems include awkward table structures, poorly defined relationships, and inconsistent field specifications; they will invariably surface later and at the least opportune times. Therefore, you should do your best to avoid this perilous situation by following the aforementioned rule. Just remember that it's always better to define a new database structure explicitly than to copy an existing structure. After all, if the old database didn't have problems, you wouldn't be building a new one.

You'll typically analyze paper-based databases and legacy databases during this part of the design process. Many organizations use both types of databases to some degree, and you perform the same basic analysis process on each of them. There are minor differences in the way you analyze a paper-based database and a legacy database, to be sure, but the differences have more to do with the databases themselves than with the overall analysis process. You needn't be concerned with these differences, however, because I've seamlessly incorporated them into the analysis process presented in this book.

Paper-Based Databases

A *paper-based database* incorporates data that is literally collected, stored, and maintained on paper, and you'll find these items in a variety of shapes, sizes, and configurations. Some of the more common formats include handwritten or printed reports and various types of preprinted forms. Anyone who has ever worked in an office for a business, organization, or government agency is very familiar with this type of database.

You'll find that analyzing a paper-based database can be a daunting task. One of your most immediate problems is finding someone who completely understands how the database works so that you can learn its use and purpose. There are several problems with the database itself, especially in terms of the way data is collected and managed. This type of database typically contains inconsistent data, erroneous data, duplicate data, redundant data, incomplete entries, and old data that should have been purged from the database long ago. Clearly, the only reason you would analyze this type of database is to identify items that you could incorporate into the new database. For example, you can extract individual pieces of data from various sections of a form in the old database and transform them into fields in the new database.

Legacy Databases

A *legacy database* is a database that has been in existence and in use for five years or more. There are several reasons that "legacy" is used as part of the name for this type of database. First, it suggests that the database has been around for a long time, possibly longer than anyone can clearly remember. Second, the word *legacy* may mean that the individual who originally created the database either has shifted responsibilities within the organization or is working for someone else and, thus, the database has become his or her legacy to the organization. Third, the term implies the disturbing possibility that no single individual completely understands the database structure or how it is implemented in the RDBMS application program.

Even if a legacy database is based on the relational model, there's no particular guarantee that the structure is sound. Unfortunately, there are many instances in which the people who created these databases didn't completely understand the concept of a relational database. (After you have read this book, you won't fall into that group.) The result is that many older databases have improper or inefficient structures.

Numerous PC-based legacy databases are improperly or inefficiently designed, too. A number of them were originally developed and implemented by folks who didn't have much experience designing databases, or were the unlucky ones who were assigned the task of creating a database for the organization and had precious little time to do so. (They had their own normal work to do.) This means that they, too, likely did not know how to take advantage of the benefits provided by the relational model. Two characteristics commonly associated with these types of databases are duplicate fields and redundant data. You'll learn later that this can cause serious problems with data integrity.

Analyzing a legacy database is somewhat easier than analyzing a paper-based database because a legacy database is typically more organized and structured than a paper-based database, the structures within the database are explicitly defined, and there is usually an application program that people use to interact with the data in the database. (The application program is valuable to you during the analysis process because it can reveal a lot of information about the data structures and the tasks performed against the data in the legacy database.) The time it will take you to perform a proper analysis will depend to some degree on the type of database, the RDBMS used to implement the legacy database, and the application program.

The key point to remember when you're analyzing either a paper-based or a legacy database is that you should proceed through the process patiently and methodically so that you can ensure a thorough and accurate analysis.

Conducting the Analysis

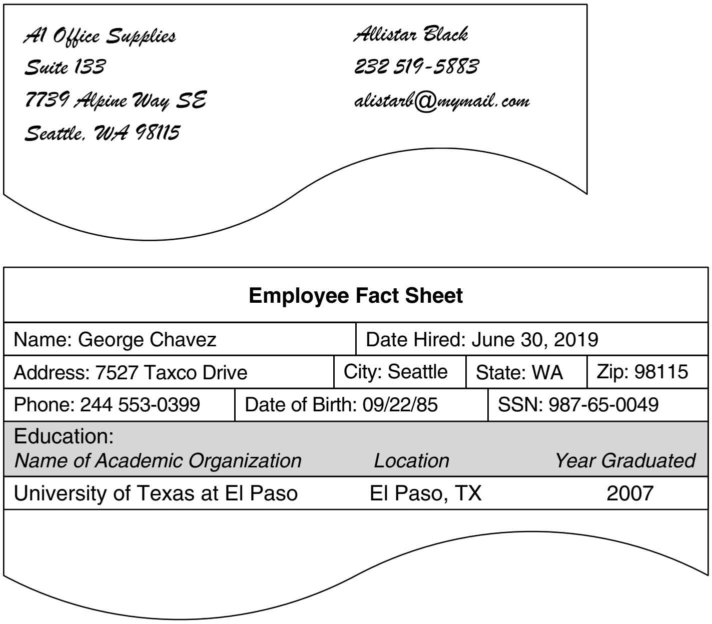
There are three steps in the analysis process: reviewing the way data is collected, reviewing the manner in which information is presented, and conducting interviews with users and management.

You will need to speak to various people in the organization as you conduct the first two steps in this process. *Be sure your conversations relate purely to the reviews at hand.* You'll have the opportunity to ask them other in-depth questions later. Keep in mind that these reviews are an integral part of your preparation for the interviews that will follow. Indeed, these reviews help you determine the types of questions you'll need to ask in subsequent interviews.

Looking at How Data Is Collected

The first step in the analysis process involves reviewing the ways in which data is collected. This includes everything from index cards and handwritten or printed lists to preprinted forms and data entry screens (such as those used in a database program or web browser).

Begin this step by reviewing all paper-based items. Find out what types of documents the organization is using to record data and then gather a single sample of each. Assemble these samples and store them in a folder for use later in the design process. For example, assume that the organization is collecting supplier data on sheets of paper in a binder. Go through the first few pages of the binder until you find one with an entry that is as complete as possible. When you've found an appropriate sample, make a copy of it and save it in the folder. Proceed through this process for each type of item being used. Figure 6.1 shows two examples of how the organization might use paper-based items to collect data.



A1 Office Supplies
Suite 133
7739 Alpine Way SE
Seattle, WA 98115

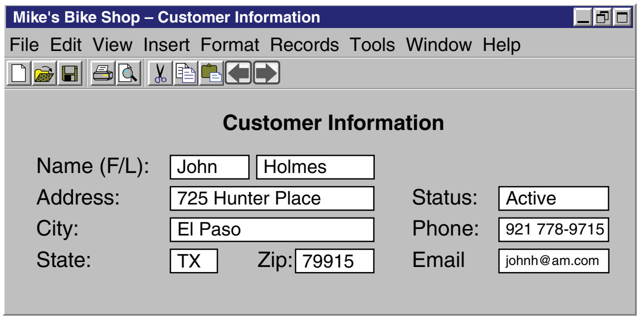
Allistar Black
232 519-5883
alistarb@mymail.com

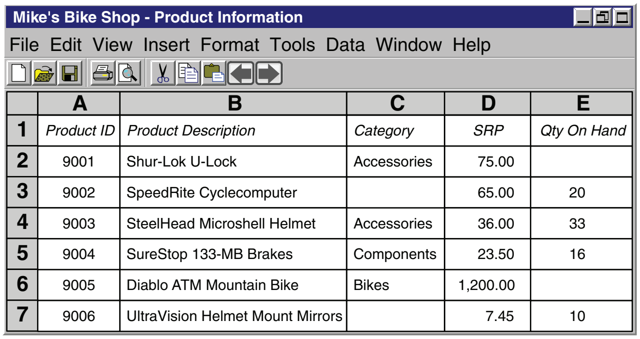
Employee Fact Sheet		
Name: George Chavez	Date Hired: June 30, 2019	
Address: 7527 Taxco Drive	City: Seattle	State: WA Zip: 98115
Phone: 244 553-0399	Date of Birth: 09/22/85	SSN: 987-65-0049
Education: *Name of Academic Organization*	*Location*	*Year Graduated*
University of Texas at El Paso	El Paso, TX	2007

Figure 6.1 *Examples of paper-based items used to collect data.*

Next, review all the computer programs that the organization uses to collect data. The objective here is to gather a set of sample screenshots that represent how the organization uses these programs to work with data. A word of caution: Many people have discovered unique and ingenious ways to use common programs, such as word processors and spreadsheets, as a way to collect and manage data. Make sure you speak with someone who is familiar with the way the computers are being used within the organization and determine which programs the organization is using to manage its data.

As you review each program, find a screen that best represents how the program collects data. You're looking for screens similar to those in Figure 6.2.





	A	B	C	D	E
1	*Product ID*	*Product Description*	*Category*	*SRP*	*Qty On Hand*
2	9001	Shur-Lok U-Lock	Accessories	75.00	
3	9002	SpeedRite Cyclecomputer		65.00	20
4	9003	SteelHead Microshell Helmet	Accessories	36.00	33
5	9004	SureStop 133-MB Brakes	Components	23.50	16
6	9005	Diablo ATM Mountain Bike	Bikes	1,200.00	
7	9006	UltraVision Helmet Mount Mirrors		7.45	10

Figure 6.2 *A typical database screen and a typical spreadsheet screen.*

The first screen is typical of those you would find in a database program, and the second screen is typical of those you would find in a spreadsheet program. When you've found an appropriate sample, create a screenshot using your favorite screen-capture application, paste it into a document in your word processing program, indicate the name of the source program and the date you created the screenshot, and then print the document. Continue reviewing the program and repeat this procedure as appropriate. Then repeat the entire process for each program. After you've printed copies of all the appropriate screenshots, assemble them together and store them in a folder for use later in the design process.

Now examine the web pages that the organization uses to collect data via the Internet. The pages you're interested in will look very similar to the data entry forms you would find in a database application program. Figure 6.3 shows an example of such a page.

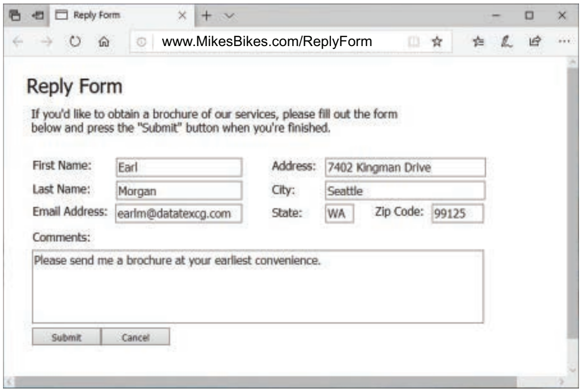


Figure 6.3 *An example of a typical online data entry screen.*

You can follow the same examination procedure here that you used with the application programs. Take a screenshot of a given web page, paste it into a word processing document, indicate the URL, the program name, and screen capture date, and print it. Continue to review the web pages and repeat this procedure as appropriate. After you've printed copies of all the appropriate screenshots, assemble them and store them in a folder for use later in the design process.

Make sure you clearly mark the folders containing the samples you've gathered during your analysis. The small amounts of time you invest to organize your materials pay big dividends when you use those materials during a complex phase of the design process.

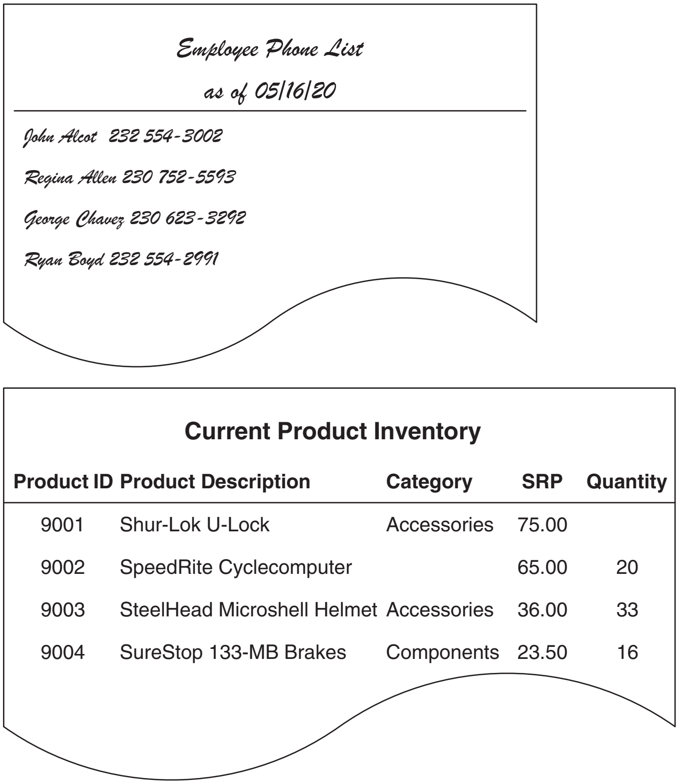
Looking at How Information Is Presented

The second step in the analysis process involves reviewing the various ways in which the organization presents its data as information. During this process, you'll review items such as handwritten documents, computer printouts, screen presentations, and web pages.

Here are three of the most popular presentation methods that you'll encounter during this process.

1. *Reports:* A report is any document (typed, printed, or computer-generated) used to arrange and present data in such a way that it is meaningful to the person or people viewing it. Although using a word processor, spreadsheet, or other software program is the standard method of generating a report, you'll still find some reports written by hand.
2. *Screen presentations (also known as slide shows):* This type of presentation incorporates a series of slides that discuss various topics in an organized manner. It is generally created with a program, such as Microsoft PowerPoint, Google Slides, or Apple Keynote, and executed on a computer.
3. *Web pages:* Many organizations have vast amounts of information available via pages on their websites. A web page is used in much the same manner as a report, and, indeed, it is really nothing more than a different *type* of report.

Begin this step by identifying and reviewing each report the organization generates from the database, regardless of whether the organization produces the report manually or with an application program. Gather samples of the reports and assemble them in a folder as you did with the items in the previous step. Overall, this task is easier to perform in this step than it was in the previous step because people in the organization are typically familiar with the reports they use. Copies of the reports are usually readily available, and most reports can be reprinted if necessary. Figure 6.4 shows an example of a report written by hand and a report generated from a word processing program.



Employee Phone List
as of 05/16/20

John Alcot 232 554-3002
Regina Allen 230 752-5593
George Chavez 230 623-3292
Ryan Boyd 232 554-2991

Current Product Inventory

Product ID	Product Description	Category	SRP	Quantity
9001	Shur-Lok U-Lock	Accessories	75.00	
9002	SpeedRite Cyclecomputer		65.00	20
9003	SteelHead Microshell Helmet	Accessories	36.00	33
9004	SureStop 133-MB Brakes	Components	23.50	16

Figure 6.4 *A handwritten report and a computer-generated report.*

Next, review screen presentations that use or incorporate the data in the database. Reviewing *every* presentation is unnecessary, but you do need to review those that have a direct bearing on the data in the database. For example, you don't need to review a presentation on the organization's new product *if it doesn't draw any data* from the database. On the other hand, a presentation on sales statistics that does incorporate data from the database is one that you do need to review.

After you've identified which presentations you need to review, go through each one carefully and make screenshots of the slides that use or incorporate data from the database. Copy the screenshots into a word processing document, print the document, and then store the document in a folder for later use. (Write the name of the presentation, its physical filename, and the date you captured the screenshots on the folder; you may need to refer to it again later.) *Follow this procedure separately for each presentation.* You want to make sure you don't accidentally combine two or more presentations together because this mistake will inevitably lead to mass confusion and result in one huge mess!

Figure 6.5 shows an example of the type of slides you'll examine during this review.

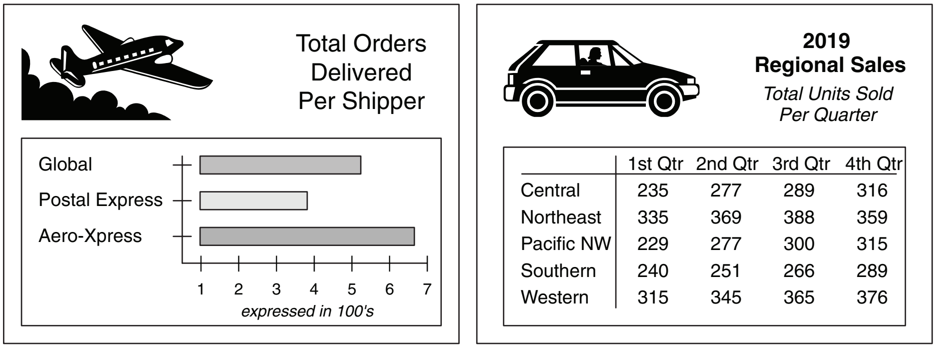


Figure 6.5 *Examples of screen presentation slides.*

Reviewing a presentation is difficult in some cases, and deciding whether you should include a given slide as a sample is purely a discretionary decision. Therefore, work closely with the person most familiar with the presentation to ensure that you include all appropriate slides in the samples.

Finally, review web pages that draw information directly from the database. Perform this review in the same manner as the review for the screen presentations. As with the previous review, you need to review those web pages that have a direct bearing on the data in the database. For example, you *don't* need to review a web page that provides a history of your organization, but you *do* need to review a web page that displays regional employee information.

After you've identified which web pages you need to review, take a screenshot of each page. Copy the screenshots into a word processing document, print the document, and then store the document in a folder for later use. (Write the URL address and the current date under each screenshot in the document; you may need to refer to a particular web page again at a later time.)

Figure 6.6 shows an example of a web page you would examine during this review.

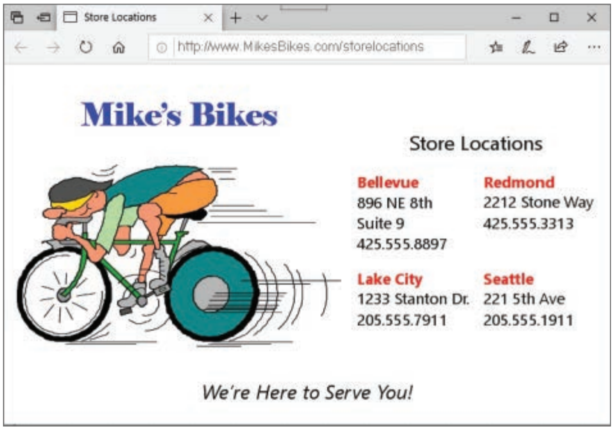


Figure 6.6 *Example of a web page that presents information from a database.*

Try to work with the person (or persons) who created and developed the organization's website. They can save you a lot of time by directing you to the exact pages you should examine for this review.

Conducting Interviews

Now that you have a general idea of how the organization collects and presents its data, it's time to interview users and management to determine how the organization *uses* its data. Interviews are useful in the analysis phase for these reasons:

- *They provide details about the samples you assembled during the previous reviews.* The discussions you had with users and management during the previous reviews were solely meant to identify (in general terms) how the organization collects and presents the data it uses. In this phase, however, you'll ask specific questions about the samples you assembled during those reviews. This will enable you to clarify the aspects of a specific sample that you consider to be vague or ambiguous.

- *They provide information on the way the organization uses its data.* These interviews will provide you with information on how users work with the organization's data on a daily basis and how management uses information based on that data to manage the organization's affairs.
- *They are instrumental in defining preliminary field and table structures.* The responses you receive from users and management during this round of interviews will help you identify initial field and table structures for the database.
- *They help to define future information requirements.* The discussions you'll have with users and management regarding the organization's future growth will often reveal new information requirements that must be supported by the database.

I cannot overemphasize, and you must not underestimate, the impact interviews have on the final database structure and how important they are to your successful completion of the database design process. Only full and complete interviews will help you ensure that the database you design fulfills your organization's information requirements.

Basic Interview Techniques

You must first learn a few basic interview techniques in order to conduct successful interviews. I address this issue here by providing you with a set of fundamental techniques that you can use to conduct every interview within the database design process. These techniques are relatively easy to learn and apply, and they'll enable you to obtain the information you require for the task.

You'll probably execute these techniques in a strict, mechanical fashion as you're just starting to learn them, but you'll apply them more instinctively and intuitively as you conduct further interviews and gain additional experience. Conducting an interview is a skill, and, as with any other skill, you will achieve various degrees of expertise with patience and practice.

The Importance of Questions

Learning how to ask a question is a valuable skill that you'll have to learn and develop if you're going to be successful at designing databases. It is what you will use to understand how your (or your client's) business works and enable you to gather the information you need to develop the various structures for the database. And, as you may have already surmised, it is precisely the skill required to conduct the interviews throughout this design process. I know this might seem like I'm stating the obvious, but I just can't overstate how important a skill this truly is.

The Interview Process

You use both open-ended and closed questions throughout an interview, alternating between each type as the interview progresses. Open-ended questions are more general in nature and enable you to focus on specific subjects, whereas closed questions are more specific and allow you to focus on particular details of a certain subject. For instance, start the interview with a few open-ended questions to establish some general subjects for discussion and then select a subject and ask more specific (closed) questions relating to that subject. You could begin by asking one of the interview participants an open-ended question such as this:

> "How would you define the work that you do on a daily basis?"

Most participants will use three or more sentences to answer this type of question. It's perfectly acceptable for a participant to provide you with a long, descriptive response because you can work with this type of response more easily than you can with one that is terse. To illustrate this point, assume the participant responds to your question in this manner:

> "As an account representative, I'm responsible for ten clients. Each of my clients makes an appointment to come into the showroom to view the merchandise we have to offer for the current season. Part of my job is to answer any questions they have about our merchandise and make recommendations regarding the most popular items. Once they make a decision on the merchandise they'd like to purchase, I write up a sales order for the client. Then I give the sales order to my assistant, who promptly fills the order and sends it to the client."

This is a very good response. The participant not only answered your question, but also provided you with the opportunity to begin asking follow-up questions. His response also suggests several subjects that you can discuss later in the interview.

> ❖ **Note** A terse response, such as "I fill out customer sales orders," provides you with little information, so you'll have to work a bit harder with the participant to get an idea of what this process involves. Terse responses commonly indicate that the participant is just nervous or uncomfortable. In this case, you could put him at ease by discussing an unrelated topic for a few moments or by allowing him to select a more familiar or comfortable subject as a starting point.

Identifying Subjects

As you ask each open-ended question, identify the subjects suggested within the response to the question. You can identify subjects by listening for the *nouns* within the sentences within the response. Subjects are always represented by nouns and identify a person, place, thing, or event (something that occurs at a given point in time). There are some nouns, however, that represent a *characteristic* of a person, place, thing, or event; you don't need to concern yourself with these just yet. Therefore, make sure you only listen and note those nouns that *specifically* represent a person, place, thing, or event. (Note that there's no need to note more than one occurrence of a given noun.) You can ensure that you account for every subject you need to discuss by listing the nouns as you identify them. In the following example, I've underlined the nouns you might pick up as you listen to the response:

> "As an account representative, I'm responsible for ten clients. Each of my clients makes an appointment to come into the showroom to view the merchandise we have to offer for the current season. Part of my job is to answer any questions they have about our merchandise and make recommendations regarding the most popular items. Once they make a decision on the merchandise they'd like to purchase, I write up a sales order for the client. Then I give the sales order to my assistant, who promptly fills the order and sends it to the client."

The nouns you've identified and listed on your sheet of paper becomes your *list of subjects*. You'll add more subjects to the list as you continue to work through the design process. Compile this list carefully and methodically because you'll use it to generate further discussions as the interview progresses and to help you define tables later in the design process.

Here are subjects (shown in alphabetical order) that are represented in the previous response:

Account Representative	Job
Appointment	Merchandise
Assistant	Sales Order
Clients	Season
Items	Showroom

You can now use this list as the basis of further questions during the interview.

> ❖ **Note** I refer to this entire procedure as the *Subject-Identification Technique* throughout the remainder of the book.

Verify that the nouns you've listed are genuine subjects by reviewing each noun and confirming that it specifically represents a person, place, thing, or event. For example, "Account Representative" does indeed represent a person, and "Appointment" is a noun that does represent an *event*; in this case, something that happens at a given point in time.

Identifying Characteristics

After you've identified the subjects suggested within the response, pick a particular subject and begin to ask follow-up questions related to that subject. You use this line of questioning to obtain as much detailed information as possible about the subject you've selected. Make sure your follow-up questions are more specific as you progress through this part of the discussion. The nature of your follow-up questions will depend on the responses you receive from the participant. Based on our sample response, for example, you could continue the discussion by asking more specific questions about sales orders, or you could begin an entirely new line of questioning regarding clients. Assume, for now, that you ask the following question to learn more about sales orders:

> "Let's discuss sales orders for a moment. What does it take to complete a sales order for a client?"

Note that this question begins with a statement directing the interview participant to focus on a particular subject. This is a technique you should use to guide your conversation after you've selected a specific subject to discuss. Also note that the question is open-ended;

it prompts the participant for details related to the subject you've selected (sales orders) and allows you to establish the focus of the participant's subsequent responses.

Now, assume that the participant gives the following reply:

> "Well, I enter all the client information first, such as the client's name, address, phone number, and email address. Then I enter the items the client wants to purchase. After I've entered all the items, I tally up the totals and I'm done. Oh, I forgot to mention I enter the client's fax number and shipping address—if they have one."

Keep the Subject-Identification Technique in mind as you listen to this response and jot down any nouns that might represent a person, place, thing, or event on your list of subjects.

With this line of questioning, though, you're going to be more interested in listening for any *details* regarding the subject under discussion. Your objective here is to obtain as many facts about the subject as possible. *Now* you're interested in nouns that represent *characteristics* of a subject. They *describe* particular aspects of that subject. You can identify these nouns quite easily because they are usually in singular form ("phone number," "address"). In contrast, nouns that identify subjects are usually in possessive form ("the *client's* phone number," "the *company's* address").

Try to listen for as many characteristics of the subject as possible. In the following example, I've underlined the nouns you might pick up as you listen to the response:

> "Well, I enter all the client information first, such as the client's <u>name</u>, <u>address</u>, <u>phone number</u>, and <u>email address</u>. Then I enter the items the client wants to purchase. After I've entered all the items, I tally up the totals and I'm done. Oh, I forgot to mention that I enter the client's <u>fax number</u> and <u>shipping address</u>—if they have one."

As you identify the appropriate nouns within a response, list them on a sheet of paper; this becomes your *list of characteristics.* You'll add more characteristics to the list as you work through the design process, and you'll use this list later when you're determining the fields for the database. *Use a separate sheet of paper for the list of characteristics. Do not list the subjects and characteristics on the same sheet!* (The reason for keeping them on different lists will become clear when you begin to define tables for the database in Chapter 7, "Establishing Table Structures.")

Here are the characteristics (shown in alphabetical order) that are represented in the previous response:

Address Name	Totals
Email Address	Phone Number
Fax Number	Shipping Address

This constitutes the list of characteristics for the subject under discussion. These characteristics will eventually become fields in the database.

❖ **Note** I refer to this entire procedure as the *Characteristic-Identification Technique* throughout the remainder of the book.

Verify that the nouns you've listed are genuine characteristics by confirming that they represent particular aspects of that subject. For example, "name" does describe a characteristic of the subject "client" and "Shipping address" does represent another characteristic of the subject "client."

After you've finished discussing a particular subject, move on to the next subject on your subjects list and begin the same pattern of questioning. Start with open-ended questions, identify any subjects suggested in the responses, ask more specific questions as the discussion progresses, and identify as many of the subject's characteristics as possible. Continue this process in a methodical manner until you've discussed every subject on your list.

You should learn the Subject-Identification Technique and the Characteristic-Identification Technique as thoroughly as possible because you'll use them during your interviews with users and management and as you identify fields and tables for the initial database structure. Note that it will become easier to distinguish subjects and characteristics in your mind as you gain experience using these techniques. They will eventually become more instinctive and intuitive.

Before You Begin the Interview Process . . .

You can use the techniques you've just learned in this section for both user interviews and management interviews. The only differences between the two sets of interviews lie in the subject matter and the content of the questions.

The interview process involves two sets of discussions: one with users and the other with management. You'll speak to the users first because they represent the "front lines" of the organization. They have the clearest picture of the details connected with the organization's daily operations. Also, the information you gather from the users should help you to understand the answers you receive from management.

Interviewing Users

The first part of the interview process involves conducting user interviews. The interviews focus on these four issues:

1. The types of data users are currently using
2. How users are currently using their data

3. The collection of samples you assembled during the first two steps of the analysis
4. And the types of information users require for their daily work

Because these issues are both data-centric and information-centric, you must be certain that you understand and always keep in mind the difference between data and information. Recall from Chapter 3, “Terminology,” that *data* are the values you store in the database and *information* is data that you process in a manner that makes it meaningful and useful to you when you work with it or view it. Keeping these definitions in mind will help ensure that you focus on each issue properly and conduct each segment of the interview successfully.

Reviewing Data Type and Usage

You can usually discuss the first two issues at the same time if you carefully phrase your questions at the beginning of the interview. Your objective for this part of the interview is to identify the types of data the users are currently using and how they use that data in support of the work they do. You’ll use this information later in the design process to help define field and table structures. Use the data collection and data representation samples to help you formulate questions about the user’s data. (However, don’t actually discuss the samples just yet; you should deal with them separately.) During this discussion, you’ll start with open-ended questions, identify subjects within the responses, and then use specific follow-up questions to identify the characteristics of each subject.

As you begin the interview, ask each participant about the work he or she performs on a daily basis. After a participant provides an overall description of the work he does, ask him to explain his job in more detail. Perhaps he can walk you through the job he performs on a daily basis.

Here's an example of a typical conversation that occurs during this part of the interview. I've underlined the nouns that should get your attention.

INTERVIEWER: "What kind of work do you do on a day-to-day basis?"

PARTICIPANT: "I accept land-use applications that are submitted by various people, log them in, and set a hearing date with the hearing examiner. I also assist applicants if they have any questions regarding a specific application."

INTERVIEWER: "Let's talk about the applications for a moment. What types of facts are associated with an application?"

PARTICIPANT: "There's quite a number, actually. There are facts concerning the type and name of the application, its designation and address, and its location."

INTERVIEWER: "Tell me about the facts concerning the application's type and name."

PARTICIPANT: "There are four things we record: the type of application, the name of the subdivision, the purpose of the project, and a description of the project."

Note how the interviewer starts the discussion with an open-ended question. As the participant responds, the interviewer uses the Subject-Identification Technique to identify subjects within the response. The interviewer then chooses a particular subject and uses another open-ended question to focus the participant's attention on that subject. Because the participant's next response is general in nature, the interviewer focuses on a particular *aspect* of the subject and uses a more specific follow-up question to elicit a detailed response from the participant.

The interviewer can continue to narrow the focus of his questions as the discussion progresses. As the participant responds to each question, the interviewer continues to use the Characteristic-Identification

Technique to identify characteristics of the subject that appear in the response. After he's identified all the subject's characteristics, the interviewer then moves on to the next subject and begins the entire process again. He'll continue in this manner until he's covered his entire list of subjects. You'll go through the same exact process when you act as interviewer.

❖ **Note** The dialogue in the previous example and in the examples throughout the book is simplistic by design; this also applies to all of the sample questions I provide during my discussion of the interview process. They are simply the vehicle through which I am presenting a specific skill or technique. As such, *don't get too preoccupied with the actual dialogue or question itself; rather, focus on how it is illustrating the skill or technique that I'm discussing at the moment.* The examples will be more beneficial to you if you see them from this perspective.

Reviewing the Samples

The next round of discussions centers on all the samples you assembled earlier in the analysis process. Your objectives during these discussions are to identify how the objects represented by the samples are used, clarify the aspects of the samples you don't understand, and assign a description to each sample.

It should be relatively easy for you to talk to participants about the samples now that you have an idea of the data the participants use on a daily basis. Begin the conversation by asking questions about a specific sample. Figure 6.7 shows an example of a data collection sample you might use as a starting point.

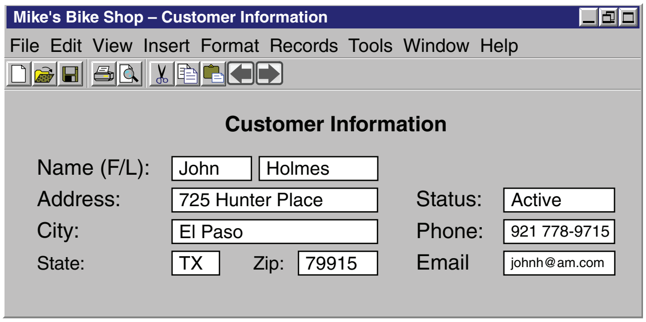


Figure 6.7 *A data collection sample.*

❖ **Note** The statement I made about the dialogue examples also applies to this figure and many of the other figures throughout the book. These figures are simply the vehicle through which I am presenting a specific skill or technique. Use the same approach that I suggested for the dialogue examples, as they'll be more beneficial to you.

Review your notes from the discussions you held at the beginning of the interview before you ask your first question. You want to determine whether anything you've *already* discussed is relevant to the sample you're about to discuss. In one of the previous discussions, for example, a participant indicated that part of his job is to keep track of all the organization's customers. Using that statement as a starting point, you could ask him how he uses this particular data collection sample to perform that task.

> "You mentioned in a previous discussion that you keep track of all the customers. How does this screen help you to carry out that task?"

This is a well-phrased question. It begins with a statement that focuses on a particular subject and then continues by bringing the participant's attention to the sample. The question is open enough to elicit a clear and complete response.

Now, assume the participant provides this response:

> "This screen allows me to enter new customers, as well as modify and maintain all the information we have on existing customers."

If this reply answers the question to your complete satisfaction, use it as the basis for a description of the sample. Otherwise, continue with an appropriate line of questioning until the participant clearly identifies the purpose and use of the sample. You must supply descriptions for all of your samples because you'll use them again later in the design process.

A sample's description should be succinct, yet clear enough to indicate the sample's purpose and how it is used. Write the description on a slip of paper and attach it to the sample. Here's an example of a description you might use for the sample in Figure 6.7:

> This screen is used to collect and maintain all customer data.

You need to understand the sample as completely as possible so that you can write a clear and concise description. If you don't understand aspects of a given sample, ask the participant to clarify them for you. For example, assume you're working with the report sample shown in Figure 6.8.

Current Product Inventory

Product ID	Product Description	Category	SRP	Quantity
9001	Shur-Lok U-Lock	Accessories	75.00	
9002	SpeedRite Cyclecomputer		65.00	20
9003	SteelHead Microshell Helmet	Accessories	36.00	33
9004	SureStop 133-MB Brakes	Components	23.50	16

Figure 6.8 *A report sample.*

If you don't know what the abbreviation "SRP" represents, make sure you have the participant clarify it for you—never make assumptions or suppositions. Doing so can waste valuable time and effort later in the process if your assumption or supposition proves to be incorrect.

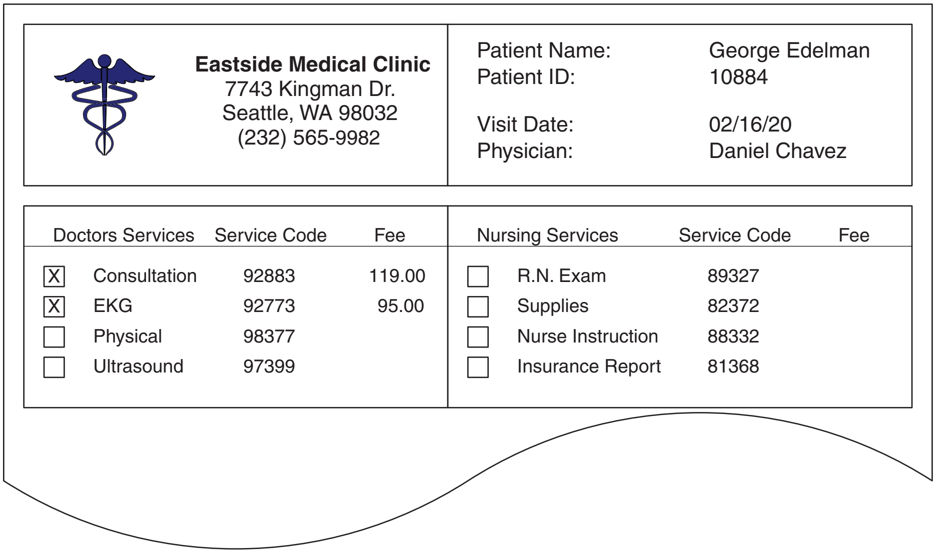
As you compose descriptions for each of the samples, you might find it difficult to write a description for a *complex* sample. A sample is complex if it represents more than one subject. The sample in Figure 6.8, for example, covers only one subject: products. The sample in Figure 6.9, however, covers at least three subjects: doctor services, nursing services, and patients. You'll often have to work a little harder to determine a complex sample's purpose and use. In some cases, you'll have to use the Subject-Identification Technique to determine what subjects are represented.

Let's say you're working with the report sample shown in Figure 6.9 and you have questions regarding the nursing services. You wonder whether the organization is using this report as an indirect means of maintaining a current list of nursing services. A question that elicits a yes or no response from a participant is not going to help you much at all, so you need to use an open-ended question that will elicit a more informative response. You could begin your discussion of this sample with this question:

> "What nursing services do you provide besides those listed in this sample?"

This type of question gives the participant an opportunity to provide you with a detailed response; furthermore, you've given yourself the opportunity to ask follow-up questions as warranted by the participant's reply. To continue the example, say you receive the following answer:

> "We provide various specialized services for the more complex patient. You see only the general services on this report. However, I can show you a complete list of our services that Katherine maintains on her computer."



Eastside Medical Clinic
7743 Kingman Dr.
Seattle, WA 98032
(232) 565-9982

Patient Name:	George Edelman
Patient ID:	10884
Visit Date:	02/16/20
Physician:	Daniel Chavez

	Doctors Services	Service Code	Fee
X	Consultation	92883	119.00
X	EKG	92773	95.00
	Physical	98377	
	Ultrasound	97399	

	Nursing Services	Service Code	Fee
	R.N. Exam	89327	
	Supplies	82372	
	Nurse Instruction	88332	
	Insurance Report	81368	

Figure 6.9 *An example of a complex report sample.*

You can continue with the process of writing the sample's description if this reply clarifies the point in question and you now understand the purpose of this report sample; otherwise, continue asking follow-up questions until everything is explained to your satisfaction.

Reviewing Information Requirements

The final issue you'll discuss with users concerns their information requirements. The objectives of this discussion are to determine whether individual users receive information based on data they don't directly control or maintain, to determine what types of additional information they need, and to determine what types of information they can foresee themselves needing in the future. You'll use the information you gather during this discussion later in the design process to help define and verify field and table structures. You can also use this information as yet another way of determining whether you accidentally overlooked anything during the previous discussions.

Current Information Requirements

Users typically receive the information they use through a variety of reports. Therefore, the best way to begin this discussion is by reviewing the report samples. This time around, though, you're not so concerned with how the reports are used as you are with the data upon which they are based. It's quite common that information on some of the reports a user receives is based on data he does not personally create and maintain. In this situation, you must determine the origin of that data so that you can identify *all* the data used by a user, whether he uses it directly or indirectly.

Select a report from the report samples and work with one of the participants to determine what data is used to produce the report. Ask him if he creates and maintains the data on which the report is based. You can move on to the next sample if he answers "yes," but you'll need to identify the origin of the data if he answers "no." Here's an example that illustrates this process.

Say you have an assistant named Kira who is beginning a discussion with a participant named Joan regarding the report sample shown in Figure 6.10.

Customer Phone List

Customer Name	Customer Type	Last Purchase	Phone Number
Alastair Black	Preferred	11/21/19	230 551-0993
Dave Cunningham	Silver	12/19/19	211 533-9182
Zachary Ehrlich	Preferred	11/16/19	211 515-3921
Bill Champlin	Gold	01/22/20	232 552-3884

Figure 6.10 *A sample report.*

As Kira begins the conversation, Joan mentions that she works in the marketing department. When Kira first asks about the sample report, Joan indicates that she receives it every Monday morning. So Kira asks her the following question:

> "Do you provide the data that's used to generate this report?"

Her next course of action depends on Joan's response. Kira can move on to the next sample if Joan's answer is yes; however, it would be a good idea for Kira to ask a follow-up question to make certain that Joan's answer is true.

> "Do you personally enter and maintain this data on a daily basis?"

If Joan's answer is still yes, Kira can definitely move on to the next sample.

On the other hand, if Joan's answer to the original question is no, Kira will need to ask a few follow-up questions. First, she'll ask Joan whether she contributes *any* data to the report. If she does, Kira will then determine what data Joan specifically submits. Then Kira will ask whether Joan knows the source of the remaining data.

To continue the example, say Joan's reply to the original question is "no" and that the following dialogue takes place after her response:

KIRA: "Can you tell me, then, if there is any data that you contribute to the report at all?"

JOAN: "I do supply the customer's name and phone number."

KIRA: "Can you tell me who provides the customer type and last purchase date?"

JOAN: "I'm not really sure, but . . ."

KIRA: "Do you have an idea of where these items come from?"

JOAN: "As a matter of fact, I do. They come from the sales department."

KIRA: "That sounds good to me. I'll make a note of that on this sample, and then we can move on to the next one."

Note that as the dialogue begins, Kira first tries to determine whether Joan submits any data at all to the report. When Joan reveals that she contributes two of the items for the report, Kira then poses a follow-up question to see if Joan can identify the source of the remaining data. In this case, it takes only two well-phrased questions to find the answer. If Joan could not answer the last two questions, Kira would need to continue her investigation with other participants.

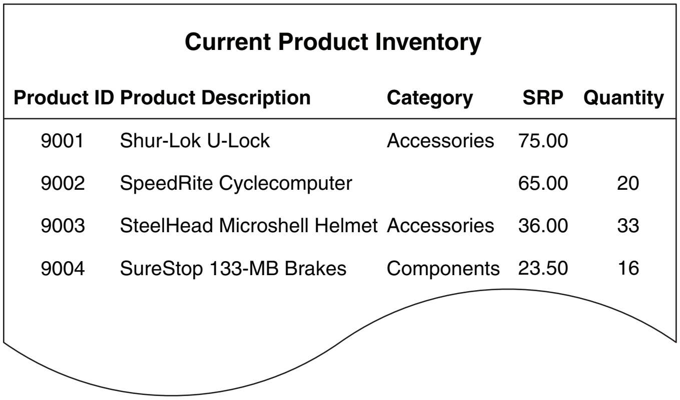
You're sure to obtain all the information you need about your report samples if your discussions progress in the same manner as the preceding dialogue. Remember: Follow-up questions are a crucial part of the conversation. You must phrase your questions properly to elicit the types of responses you need from the participants.

Additional Information Requirements

The next subject of discussion is *additional* information requirements. The objective here is to determine whether users require additional information that is not being delivered to them currently. If this is the case, you must identify what additional information they require and then define new data structures to support this extra information later in the design process.

Start this conversation by directing the participants to review the reports they currently receive. Ask them whether there is other information they would like to see in their reports. Next, direct them to discuss the additional information, which reports the information will affect, and the reason they believe the information is necessary. Then determine whether the additional information represents new subjects or new characteristics. If it does, identify each new item and add it to the appropriate list. Finally, review the participants' comments and determine whether there are further issues you need to discuss with them in regard to the reports. Here's an example that illustrates the process.

Say you're beginning this discussion and you've just asked the participants to review the report samples they currently use. One of the participants is reviewing the sample report shown in Figure 6.11.



Current Product Inventory

Product ID	Product Description	Category	SRP	Quantity
9001	Shur-Lok U-Lock	Accessories	75.00	
9002	SpeedRite Cyclecomputer		65.00	20
9003	SteelHead Microshell Helmet	Accessories	36.00	33
9004	SureStop 133-MB Brakes	Components	23.50	16

Figure 6.11 *The sample report being reviewed by a participant.*

You now instruct this participant to note the additional information she would like to see on the report and to provide a brief statement indicating why the information is necessary. It doesn't really matter exactly *how* she makes the notations as long as they are clear and attached to the report in an obvious manner. In this case, she decides to use large sticky notes as a means of documenting her comments. She's specified two new fields she would like to add to the report, along with the reason for their inclusion. She's also suggested possible locations for the fields by writing their names on the report itself. Figure 6.12 shows the sample report with her comments.

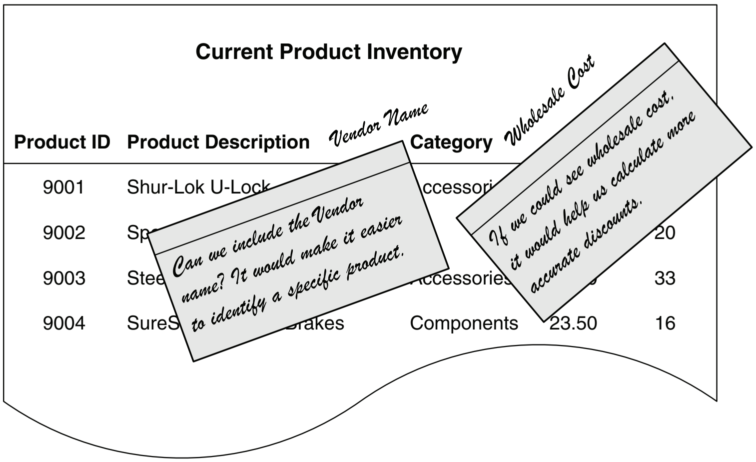


Figure 6.12 *A report sample with a participant's comments.*

Next, determine whether there are *new* subjects or *new* characteristics represented in the additional information. Keep the Subject-Identification Technique and the Characteristic-Identification Technique in mind as you refer to the comments attached to the report. Here's an example of how you apply these techniques to the first comment in Figure 6.12.

❖ **Note** From this point forward, I will always underline the nouns in an example that should get your attention, when applicable.

> "Can we include the vendor name? It would make it easier to identify a specific product."

Here you've identified both a subject and a characteristic. (Note that the subject and characteristic aren't directly related: "vendor name" is a characteristic of a vendor, not of a product. There's no problem here, but you should be aware that this apparent mismatch of subjects and characteristics is typical. You'll address this issue later in the design process.) Now, check your subjects list and characteristics list to determine whether you've already accounted for these items. If you have, move on to the next comment and repeat this procedure.

If you do discover a new subject, add it to your list of subjects and then identify as many of its characteristics as possible. When you're finished, add these items to your list of characteristics, move on to the next comment, and repeat the entire procedure. In many instances, however, you'll only identify new characteristics. Don't be alarmed. People often want to add items to a report that are characteristics of subjects that are *already* represented by the information on the report.

Finally, re-examine each report and determine whether you have questions or concerns about the notes participants have made. For instance, you may question the rationale behind one participant's belief that specific fields are necessary on a given report. Or you might wonder why another participant wants to *exclude* certain fields from one of his reports. You definitely want to make sure that the fields he wants to exclude are truly unnecessary and that removing them will not have an adverse effect on the information the report provides to other people. In either case, the inclusion or exclusion of fields will affect the final database structure.

If a report has one or more remarks that are cause for concern, review it with the appropriate participant and settle as many of the issues as you can. You can usually resolve all your concerns with a few simple questions, but in some cases the resolution to certain issues will not become apparent until later in the design process. For example, you might have noticed that certain fields appear on two or more reports. Determining whether the fields are being unnecessarily duplicated is difficult to do until you begin to define the field and table structures. When you encounter an issue that is difficult to resolve at the present time, make a note of it and put the report aside for later review.

Future Information Requirements

The last subject of discussion concerns *future* information requirements. Your objective here is to identify the information that the participants believe will be necessary for them to receive as the organization evolves. After you identify these future information requirements, you can ensure that you define the data structures necessary to support that information.

You first need to make sure that every participant has some idea of how the organization is evolving. The nature of the organization's evolution will determine what new information participants will require. If several people are unacquainted with these issues, you'll need to obtain this information from management and then relay it to the participants prior to the discussion. You can begin the conversation after everyone is familiar with these matters.

Start the discussion by directing the participants to think about the future evolution of the organization and how it may affect the work they do on a daily basis. You'll often find that some participants are going to have a difficult time envisioning this scenario. When this happens, use questions such as these to help them focus their thoughts:

> How will the organization's evolution affect the amount of information you'll need to do your job?
>
> Do you think you'll need additional types of information to carry out your duties effectively as the organization evolves?
>
> How will the evolution of the organization increase the time you spend on your daily tasks?
>
> Can you predict what types (categories, not specific items) of new information you'll need to carry out your duties as the organization evolves?
>
> Do you anticipate a need for new information if your duties are increased as a result of the organization's evolution?

Keep in mind that most of the participants' answers will be based on *speculation.* There's no accurate way for them to predict what types of information they'll really need until the organization's evolution occurs. However, if you can anticipate their hypothetical information requirements, you can prepare for them by defining the necessary data structures in advance.

As the participants respond, keep the Subject-Identification Technique in mind so that you can identify any new subjects and then add them to your list of subjects. Keep the Characteristic-Identification Technique in mind so that you can listen for any new details concerning existing or new subjects and add them to your list of characteristics.

You can sketch ideas for new reports or data entry forms to help participants visualize the types of information they may need in the future. These sketches can then help you identify new subjects or characteristics that the database structure needs to address. If you create several rough drawings of sample reports, be sure to assemble them in a separate, clearly marked folder. Then code each revision so that you can compare it with earlier revisions. Figure 6.13 shows an example of a preliminary design for a future report.

Continue the conversation with users until you're satisfied that you've accounted for as many of the participants' future information requirements as possible. When you've completed the discussion, you're ready to conduct interviews with management.

1st Quarter Customer Sales Statistics

		Sales Amounts		
Customer ID	Customer Name	Maximum	Minimum	Average
9001	Stewart Jameson	265.00	23.00	55.00
9002	Shannon Black	550.00	125.00	70.00
9003	Estela Rosales	250.00	35.00	36.00
9004	Bill Chmplin	325.00	20.00	25.00

Figure 6.13 *An example of a design for a new report.*

> ❖ **Note** You can use all the techniques you learned in this section for the management interviews as well. Therefore, the next section is somewhat shorter and more concise.

Interviewing Management

The second part of the interview process involves interviewing management personnel. This round of interviews focuses on these issues:

1. The types of information managers currently receive
2. The types of additional information they need to receive

3. The types of information they foresee themselves needing
4. Their perception of the organization's *overall* information requirements

❖ **Note** Throughout the remainder of the book, I use the term *management* to refer to the person or persons controlling or directing the organization.

Reviewing Current Information Requirements

Your objectives during the first part of this interview are to identify the information that management routinely receives and to determine whether it receives reports that are not represented in your group of report samples.

As you begin the interview, ask participants about the work they perform and the responsibilities associated with the position. Managers typically have a number of issues on their minds, so these questions will help focus attention on the matters at hand. The answers will give you some idea of how managers might use the information on the reports they receive and will provide you with a perspective on the need for that information.

Next, ask participants if they use any of the reports in your collection of report samples. Proceed with the next step if they say they don't use any of the reports; otherwise, examine each report and ask them to help you identify other subjects that you might have previously overlooked. Keep the Subject-Identification Technique in mind to aid you in this process. If the manager identifies a new subject, add it to your list of subjects and apply the Characteristic-Identification Technique to determine the subject's characteristics. Add the new characteristics to your list of characteristics. Repeat this entire procedure for each sample report.

Continue the discussion by asking participants whether they receive reports that are *not* represented in your report samples. If the answer is "yes," obtain a sample of each new report and review it with the participant. Again, keep the Subject-Identification Technique and the Characteristic-Identification Technique in mind to identify the subjects (and their associated characteristics) represented within the report, and then add the subjects and characteristics to their respective lists. Finally, attach a description to the report and add it to your collection of report samples. Repeat this procedure until you've accounted for every new report.

Reviewing Additional Information Requirements

The next subject of discussion concerns management's need for *additional* information. Your objective is to determine whether it requires supplemental information that is currently missing from the reports they receive. If you conclude that this is the case, you must identify that additional information. You'll then define new data structures (as appropriate) to support this information later in the design process. However, you can move on to the next part of the interview if management doesn't require additional information.

You use the same techniques for this discussion as those you used for this segment of the user interviews. Here are the steps you'll follow.

1. Review the report samples with the participants once again and ask them if there is additional information they would like to include in any of the reports.
2. Have the participants note the additional information, including the reasons that they believe it's necessary, on the appropriate reports. Remember that it doesn't matter how the participants make the notations as long as they are clear, noticeable, and attached to the appropriate report.

3. Identify new subjects or characteristics within the information and add them to the appropriate list.
4. Review the reports and discuss any concerns you have about them with the participants. After your concerns are resolved, this process is complete.

Reviewing Future Information Requirements

Future information requirements are the next subject of discussion. Your objective here is to determine what information management foresees itself needing in the future. After you've identified these requirements, you can ensure that data structures are in place to support this information as the need for it arises.

As you begin the discussion, have the participants consider how the organization is currently evolving. Then ask them how this evolution will affect the information they require to make sound decisions and how it will influence the way they guide or direct the organization. Remember that their answers are going to be based on speculation, as was the case with the similar questions you asked users; there's no way for management to predict its future needs accurately until the organization actually begins to evolve. (It's always a good idea, however, to plan for the future as much as possible.) Keep the Subject-Identification Technique and Characteristic-Identification Technique in mind to identify new subjects and characteristics within the participants' responses and then add the new items (if any) to the appropriate lists.

Next, make sketches of any new reports the participants might have in mind. Identify new subjects and characteristics within each report and add them to the appropriate lists. Assemble these new reports in a clearly marked folder and add it to your collection of samples.

You're ready to move on to the last subject when you've accounted for as many of management's future information requirements as possible.

Reviewing Overall Information Requirements

The last topic of discussion concerns the organization's *overall* information requirements. In management's opinion, what generic class of information does the organization need? Your objective here is to discover whether there is data that the organization needs to maintain that has not been previously discussed in either the user interviews or the management interviews. If you determine that there is such data, you must account for it in the database structure.

Take all the reports that you've gathered throughout the analysis and interview processes and review them with the participants once more. Ask the participants to consider the information the reports provide and how they might use that information. (Note that they'll have to make assumptions about how they might use the information from the new reports.) Next, ask participants to determine whether there is information that would be useful or valuable to the organization, but that is not currently being received by *anyone* within the organization. If they determine that there is, indeed, some new information that the organization could use, go through the normal process of identifying that information and the subjects and characteristics represented within it. Sketch samples of new reports for the information, as appropriate, and add the samples to your existing collection of new reports.

For example, assume that one of the participants has identified a need for demographic information; she believes that it would help the organization identify a more specific target market for its product. None of the existing reports furnishes this information, so you identify exactly what she needs by working with her to create a sketch of a report that will present this information. (She might actually sketch more than one report, but this is neither a problem nor a cause for concern.) You then use the appropriate techniques to identify and note the subjects and characteristics represented within the report and add it to your existing collection of new reports. Later in the design process, you'll define the data structures necessary to support the new information.

Repeat this procedure until the participants can no longer identify any further information that the organization might find useful or valuable. After you're reasonably confident that you've accounted for all the organization's information requirements, suspend the interview process and begin the process of compiling the Preliminary Field List.

It's important for you to understand that you may have to revisit this process, even though you and the participants may believe that you've accounted for all the information the organization could possibly use. You'll commonly identify new information as the database design process unfolds.

Compiling a Complete List of Fields

The Preliminary Field List

Now that you have completed your analysis of the current database and the interviews with users and management, you can create a *Preliminary Field List.* This list represents the organization's fundamental data requirements and constitutes the core set of fields that you'll define in the database. You create the Preliminary Field List using a two-step process.

Step 1: Review and Refine the List of Characteristics

The first step involves reviewing and refining the list of characteristics you compiled throughout the analysis and interview process. As you learned in Chapter 3, a *field* represents a characteristic of a particular subject; therefore, each item on your list of characteristics will become a field. Before you transform those characteristics into fields, however, you first need to review the list to identify and remove duplicate characteristics.

During the interviews, you identified various characteristics within each participant's responses and compiled them into a list as the interview progressed. There were probably times when you mistakenly added the same characteristic to the list more than once, or unknowingly referred to the *same* characteristic by two or more different names. As a result, your list of characteristics requires some refinement.

Refining Items with the Same Name

Begin refining your list of characteristics by looking for items with the same name. When you find one or more occurrences of a particular name, determine whether they all represent the same characteristic. Cross out all but one occurrence of the name from the list if they do represent the same characteristic; otherwise, determine what each instance of the name represents. You'll often find that a duplicate name represents the *same type* of characteristic as its original counterpart but should be associated with a different subject than its counterpart. In this case, you rename the duplicate to reflect how it relates to the appropriate subject.

Assume, for example, that the item "Name" appears three times on your list of characteristics. Your first inclination will probably be to cross out two of the occurrences because your current objective is to eliminate duplicate characteristics. However, you should determine whether each instance of "Name" represents a distinct characteristic before you remove it. You can easily make this determination by examining your interview notes; this will help you remember when and why you added the item to the list.

After careful examination, you discover that the first occurrence of "Name" represents a characteristic of the subject "Clients," the second, a characteristic of the subject "Employees," and the third, a characteristic of the subject "Contacts." You resolve this duplication by *renaming* each occurrence of "Name" (using the subject as a prefix) to reflect its true meaning. Now you'll have three new characteristics called "Client Name," "Employee Name," and "Contact Name."

Items similar to "Name" commonly appear on a list of characteristics, and you must address them in the same manner. You'll commonly see one or more occurrences of items such as "Address," "City," "State," "ZIP Code," "Phone Number," and "Email Address," and you can refer to them collectively as *generic items.* The point here is that you must rename each instance of a generic item to reflect its true relationship to a particular subject, thus ensuring that you have as accurate a field list as possible.

Refining Items Representing the Same Characteristic

Now look for items that represent the *same* characteristic and cross out all but one. The idea here is that a given characteristic should appear only once in the list of characteristics. For example, assume that "Product #," "Product No.," and "Product Number" appear on your list of characteristics. It's evident that these items all represent the same characteristic and you need only one of them on your list. Choose the one that conveys the intended meaning clearly, completely, and unambiguously and cross out the remaining items from the list of characteristics. (In this case, the best choice is "Product Number" because it fulfills the previous criteria.)

Ensuring Items Represent Characteristics

Finally, make sure that each item on your list represents a *characteristic.* It's easy to place items accidentally on the list that represent subjects. You can test each item by asking yourself questions such as these:

> Can this word be used to describe something?
>
> Does this word represent a component, detail, or piece of something in particular?

> Does this word represent a *collection* of things?
>
> Does this word represent something that can be broken down into smaller pieces?

Depending on the item you're working with, some questions are easier to answer than others. When you find that an item represents a subject rather than a characteristic, remove it from the list of characteristics and add it to the list of subjects. Be sure to identify the new subject's characteristics and add them to the existing list of characteristics.

For example, say "Item" appears on your list of characteristics and you're not quite sure whether it represents a characteristic or a subject. Use the preceding questions to help you make a determination.

> Can "Item" be used to describe something?
>
> Does "Item" represent a component, detail, or piece of something in particular?

You could make a case that "Item" helps to describe a sale inasmuch as it identifies what a customer purchased. On the other hand, you could also say that "Item" isn't a characteristic because it doesn't represent a *singular* aspect of a sale. "Date Sold," for example, represents a singular characteristic of a sale. Leaving the quandary surrounding these questions unresolved, you go on to the next question:

> Does "Item" represent a collection of things?

You can answer this question easily by looking at the plural form of the word, which in this case is "Items." If "Items" can be referred to as a collection, it *is* a subject. It's beginning to become clear that "Item" does represent a collection of some sort, and you can make a final determination by asking yourself the last question:

> Does "Items" represent something that can be broken down into smaller pieces?

You can answer this question by determining whether you can identify any characteristics for “Items.” If you can, then “Items” definitely represents a subject, and you should move it to the list of subjects. You also need to identify its characteristics and add them to your list of characteristics.

Continue with this procedure until you’ve reviewed and refined the entire list of characteristics to your satisfaction. When you are through, you have your first version of the *Preliminary Field List.* Now you’ll add new items to it and refine it further during the next step.

Step 2: Determine Whether There Are New Characteristics in Any of Your Samples

This step involves an examination of all the samples you gathered throughout the analysis process. Your goal is to determine whether there are characteristics on the samples that need to be added to the Preliminary Field List.

Begin this step by highlighting every characteristic you find on each sample. Then, examine each characteristic and determine whether it’s already on the Preliminary Field List; cross it out on the sample if it’s already on the list. Next, study the remaining characteristics and determine whether any of them has the *same meaning* as an existing field; if it does, cross it out on the sample. (Use the same procedure you used in the first step to make this determination.) Finally, add any highlighted characteristics remaining on the samples to the Preliminary Field List.

For example, say you’re working with the data collection sample shown in Figure 6.14.

Highlight each characteristic you find on the sample, as shown in Figure 6.15.

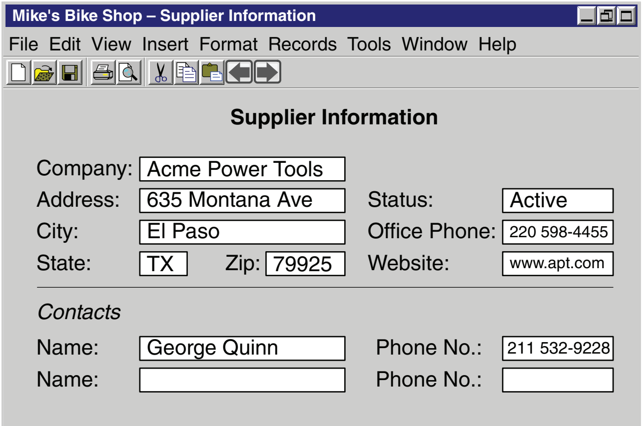


Figure 6.14 *An example of a data collection sample.*

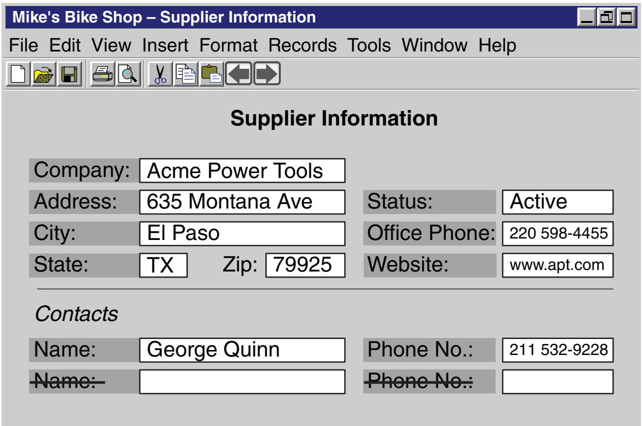


Figure 6.15 *A sample with highlighted characteristics.*

You're likely to find multiple occurrences of various characteristics in some of the samples. As you can see, both "Name" and "Phone No." appear twice on this particular sample. You can cross out the duplicates in this case because they have the same meaning as the original instances.

To continue with the example, say you reviewed the Preliminary Field List and found that every characteristic on the sample is already on the list with the exception of "Name" and "Phone No." Cross out the existing items on the sample to show that you have accounted for them. Before you add "Name" and "Phone No." to the Preliminary Field List, however, make sure that the names of these items properly describe their relationship to the subject represented within the sample. In this case, the two remaining items represent characteristics of a group of people known as "Contacts." Therefore, you rename these characteristics, using the subject as a prefix, as "Contact Name" and "Contact Phone Number," and then add them to the Preliminary Field List. Repeat this procedure for each sample you've gathered until you've gone through all the samples you've collected. When you're through, you have the *second* version of the Preliminary Field List.

A Side Note: Value Lists

As you examine the characteristics on a database, spreadsheet, or web page sample, record on a separate sheet of paper the name of each characteristic that incorporates a *value list* (also known as an *enumerated list*). This list specifies the acceptable range of values for a particular characteristic and often enforces a given business rule. (You'll learn about business rules in Chapter 11, "Business Rules.") For example, say you work for a manufacturing company that uses four specific vendors to deliver its goods to customers across the nation. You could use a value list to ensure that a user selects one of those four vendors to ship a particular order. Figure 6.16 illustrates this example (note Ship Via) and also shows two common types of value list.

When you record the name of a characteristic that incorporates a value list, also record the values within the list. If the list contains a large number of values, write a brief description of the type of values in the list and (if possible) a minimum and maximum value; otherwise, write down each of the values. Figure 6.17 shows an example of the record you're creating.

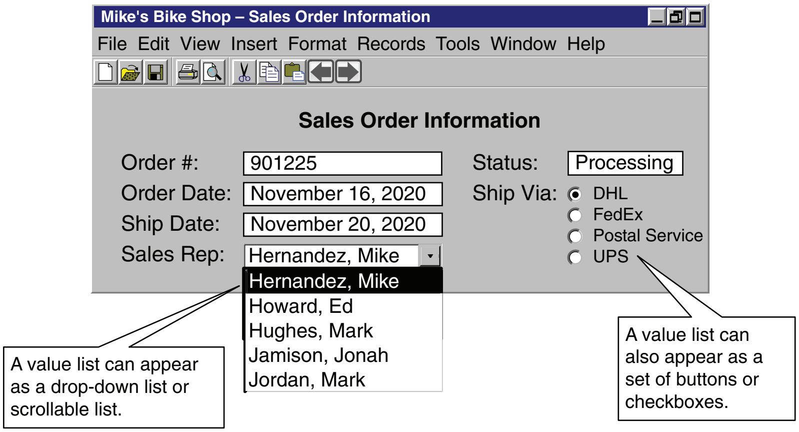


Figure 6.16 *A database screen with two value lists.*

Characteristics Incorporating a Value List

Characteristic	Value List
Category	Accessories, Bikes, Clothing, Components, Maintenance, Racks, Wheels
Department	Accessories, Bikes, Clothing, Service
Sales Rep	The name of every employee within the organization whose position is that of a sales rep
Ship Via	DHL, FedEx, Postal Service, UPS

Figure 6.17 *Recording characteristics that incorporate value lists.*

You can be discerning about the characteristics you choose to record. For example, you don't need to record characteristics that accept simple or obvious sets of values, such as "yes/no," "true/false," or "active/inactive." Instead, you should record characteristics that accept distinct, specific sets of values.

Set this sheet (or sheets) aside after you've finished recording the appropriate characteristics. You'll refer to this sheet when you define field specifications for the fields in the database and again when you define business rules.

The Calculated Field List

There's one final refinement you must make to the Preliminary Field List before you can consider it complete: You must remove every *calculated field* and place it on a separate list. This new list becomes your *Calculated Field List.* Recall from Chapter 3 that a calculated field is one that stores the result of a string concatenation or mathematical expression as its value. You list calculated fields separately because you'll use them in a specific manner later in the design process.

You build the Calculated Field List using existing fields from the Preliminary Field List. Examine the list and determine whether there are fields that fit the description of a calculated field. Fields that have names containing words such as *amount, total, sum, average, minimum, maximum,* and *count* are likely candidates for the Calculated Field List. Common names for calculated fields include "Subtotal," "Average Age," "Discount Amount," and "Customer Count." As you identify each calculated field, remove it from the Preliminary Field List and place it in the Calculated Field List. When you've completed your examination of all the fields in the Preliminary Field List, you'll have two completely new lists: a *third* version of the Preliminary Field List and a Calculated Field List.

Reviewing Both Lists with Users and Management

Conduct brief interviews with users and management to review the items that appear on the Preliminary Field List and the Calculated

Field List. Your objective here is to determine whether any fields have been omitted from either list. You can continue with the next step in the design process when everyone is satisfied that the lists are complete; otherwise, identify the fields that are missing and add them to the appropriate list. After the interviews are complete, you'll have a "final" version of each list.

Be sure you conduct these interviews because the participants' feedback provides you with a means of verifying the fields on both lists. Let me remind you once again to avoid becoming too invested in the idea that these lists are absolutely complete and final. At this point you still may not have identified every field that needs to be included in the database (inadvertently; you're almost sure to miss a few fields), but if you strive to make your lists as complete as you can, the inevitable additions or deletions will be quick and easy to make.

EXAMPLE: ANALYZING A DATABASE

Let's work with Mike and his new database once again. You've already defined the mission statement and mission objectives for Mike's new database. Now it's time to perform an analysis, conduct interviews, and compile a Preliminary Field List.

First, analyze Mike's current database. You find that he keeps most of his data on paper; the only exception is the product inventory he maintains in a spreadsheet program. Gather samples of the various papers Mike uses to collect data and a screenshot or printout of the spreadsheet he uses to maintain the product inventory. Assemble these samples together in a folder for later use. For example, Figure 6.18 shows a sample of the lists Mike uses to collect customer information, along with a screenshot of his spreadsheet program.

Steven Horst 232 363-9755

Apartment 2B
2380 Redbird Lane
Seattle, WA 98115

He's primarily interested in mountain bike stuff.
Keep him abreast of the summer bike tours.

Mike's Bike Shop - Product Information

File Edit View Insert Format Tools Data Window Help

	A	B	C	D	E
1	*Product ID*	*Product Description*	*Category*	*SRP*	*Qty On Hand*
2	9001	Shur-Lok U-Lock	Accessories	75.00	
3	9002	SpeedRite Cyclecomputer		65.00	20
4	9003	SteelHead Microshell Helmet	Accessories	36.00	33
5	9004	SureStop 133-MB Brakes	Components	23.50	16
6	9005	Diablo ATM Mountain Bike	Bikes	1,200.00	
7	9006	UltraVision Helmet Mount Mirrors		7.45	10

Figure 6.18 *A paper-based and a computer-generated sample from Mike's Bikes.*

Next, identify the methods Mike uses to present information. He and his staff currently produce a variety of reports that present the information they need to conduct their daily affairs. They generate most of the reports using a word processing program. Gather samples of all the reports and place them in a folder for later use. Figure 6.19 shows a sample report that Mike creates on his computer.

Supplier Phone List

Company Name	Contact Name	Phone Number
ACME Cycle Supplies	George Chavez	230 633-9910
B & M Bike Supplies	Carol Ortner	230 527-3817
CycleWorks	Julia Black	221 527-0019
Evanstone's Cycle Warehouse	Allan Davis	211 636-9360

Figure 6.19 *A report sample from Mike's Bikes.*

Now you're ready to interview Mike's staff. Here are some points to remember as you're conducting the interviews.

- *Identify the types of data staff members are using and how they use that data.* Be sure to use the Subject-Identification Technique and the Characteristic-Identification Technique to help you analyze responses and formulate follow-up questions.
- *Review all the samples you gathered during the beginning of the analysis process.* Determine how each sample is used, write an appropriate description, and attach the description to the sample.
- *Identify the staff's information requirements.* Determine what information they're currently using, what additional information they need (remember to use the samples), and what kind of information they believe they'll need as the business evolves.

During the interview, one of the employees wonders whether she can add a new field to the supplier phone list report. How do you respond? You hand her the report and ask her to attach a note indicating the name of the new field and a brief explanation of why she believes it's necessary. When she's finished, return the sample to the report samples folder. Figure 6.20 shows the report sample with the attached note.

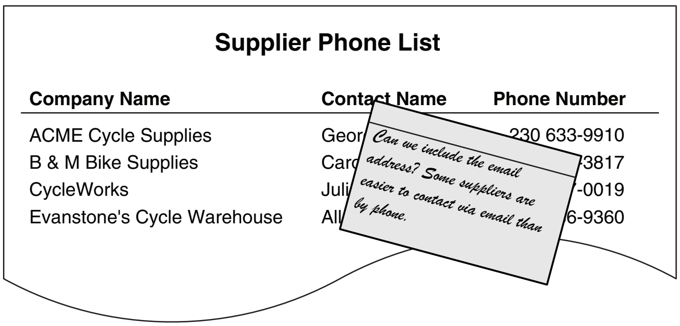


Figure 6.20 *A report sample with attached note suggesting a new field.*

You'll conduct the final interview with Mike. Keep the following points in mind as you speak with him.

- *Identify the reports he currently receives; you need to know what kind of information he uses to make business decisions.* If he receives reports that are not represented in your group of report samples, obtain a sample of each report and add it to the group, updating the subject and characteristic lists as needed.
- *Review the group of report samples with him and determine whether he can identify subjects or characteristics that his staff overlooked.* Use the appropriate techniques to identify these items and then add them to the appropriate list.
- *Determine whether there is any additional information Mike needs to supplement the information he currently receives.*
- *Determine what types of information Mike will need as the business evolves.*

As you and Mike discuss his future information needs, he indicates that there is some new information he would like to receive after the business really gets rolling: He would like to see total bike sales by manufacturer. He believes this information would help him determine which bikes he should consistently keep in stock. Such a report does not currently exist, so have Mike sketch it out on a sheet of paper. Next, identify the subjects and characteristics represented within the report and add them to the appropriate list. Add the new report to your group of report samples. Figure 6.21 shows the sketch of Mike's new report.

Your analysis is now complete. You've interviewed Mike and his staff, you've gathered all the relevant samples, and you've created a list of subjects and a list of characteristics. A *partial* list of subjects and characteristics is shown in Figure 6.22. All you need to do now is to create your Preliminary Field List.

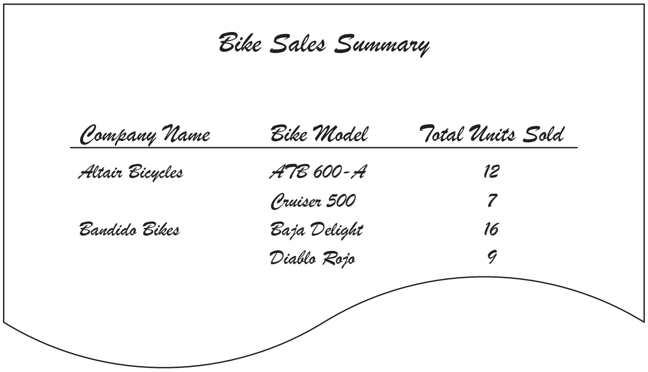


Figure 6.21 *The sketch of Mike's new report.*

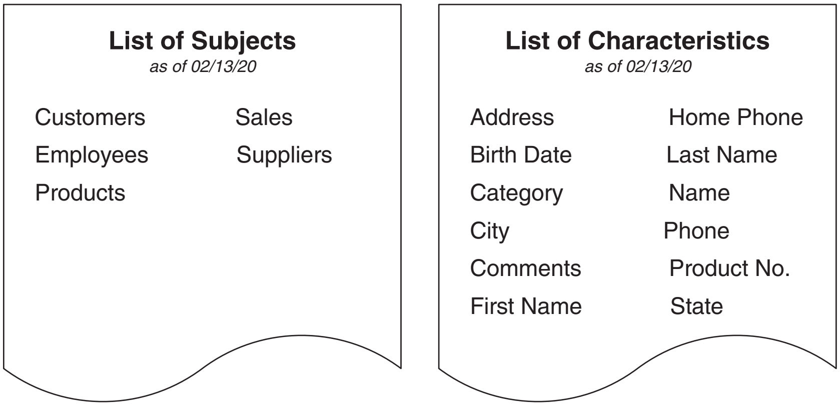


Figure 6.22 *Partial lists of subjects and characteristics for Mike's Bikes.*

As you already know, you need to refine the list of characteristics before it can become the first version of the Preliminary Field List. Cross out duplicate characteristics, the items that represent the *same* characteristic, and refine those items that have generic names. (Remember the problem with the characteristic called "Name"? If you find such characteristics, now is the time to resolve them.) Next, review all your samples and determine whether they contain characteristics that do not currently appear on the Preliminary Field List. Add to the list any new characteristics that you find. When you complete these tasks, you have the first version of your Preliminary Field List.

Now you cross out all the calculated fields from the Preliminary Field List and place them on their own list; this becomes your new Calculated Field List. Figure 6.23 shows a *small portion* of your final Preliminary Field List and Calculated Field List.

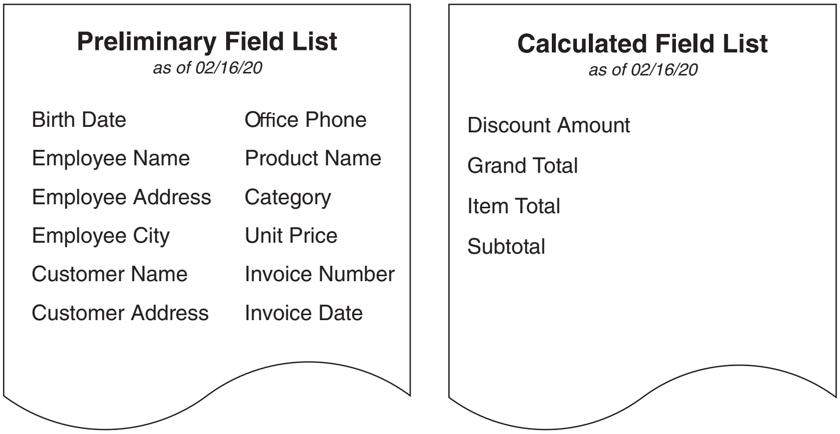


Figure 6.23 *A partial Preliminary Field List and a Calculated Field List.*

> ❖ **Note** You may have noticed that each list includes a date in the title. Dating your lists is a good idea so that you can maintain a clear history of their development.

Summary

This chapter began by discussing why you should analyze the organization's current database. You learned that the analysis helps you identify aspects of the current database that will be useful to you when you design the new database. Armed with this information, you can design a database that best suits the organization's needs. Next, we briefly looked at the two types of databases organizations commonly used: *paper-based* databases and *legacy* databases. We ended this discussion by identifying the *three steps used in the analysis process:* reviewing the way data is collected, reviewing the way information is presented, and conducting interviews with the organization's staff.

The chapter continued with a discussion of the review process. You learned how to review the ways the organization collects its data and how to assemble a set of *data collection* samples. You then learned how to *review the ways the organization presents information* and how to assemble a set of *report* samples.

Next, we discussed the process you use to *conduct interviews,* and you learned why interviews are useful at this stage of the design process. During this discussion you learned two techniques that are crucial to the success of interviews: the *Subject-Identification Technique* and the *Characteristic-Identification Technique.*

Conducting *user interviews* was the next subject of discussion. We examined the four issues you must address during these interviews, along with the techniques you use to address them. Next, we discussed conducting *management interviews.* Here you learned about the issues and techniques these interviews incorporate.

Finally, we discussed the process of *compiling a list of fields* based on the list of characteristics and the characteristics that appear in the samples. You learned that you decompose the field list into two separate lists: a *Preliminary Field List* and a *Calculated Field List.* The Preliminary Field List enumerates the organization's fundamental data requirements and establishes the core set of fields you must define in the database. The Calculated Field List consists of fields that contain values resulting from string concatenations or mathematical expressions. You also learned how to compile a value list, which is a list that specifies the acceptable range of values for a particular field in the Preliminary Field List and often helps to enforce a given business rule.

Review Questions

1. State two goals of analyzing the current database.
2. True or False: You can adopt the current database structure as the basis for the new structure.
3. What is a *legacy database?*
4. State two steps of the analysis process.
5. Which types of computer software programs should you review during the analysis?
6. Why should you conduct interviews after you gather data collection and information presentation samples?
7. How do you use "open-ended" and "closed" questions?
8. What is the *Subject-Identification Technique?*
9. How do you identify specific attributes for a particular subject?
10. True or False: You should interview users and management at the same time.
11. What three basic types of information requirements must you identify?
12. What is the *Preliminary Field List?*
13. State why each item on the Preliminary Field List should have a unique name.
14. What is a *value list?*
15. What are *calculated fields?* What (if anything) should you do about them?